

PROJECT: Propane Dehydrogenation Project
Location: Strathcona County, AB
Title: U/G Tank Buoyancy Verification
Calculation No.:
By: Coltin Walsh

Contact No.: A6KV
Date: 12-Oct-2021

Check Level: 2B
Chk'd: T. Hill

REVIEWED

By Tara Hill at 12:57 pm, Oct 12, 2021

1.0 Caustic Tank 10-TA-18005 Buoyancy Verification

As per site correspondence, the water level for 10-TA-18005 is higher than indicated on the Vendor GA drawings.

Good Afternoon Everyone,

The hydrovacating was completed yesterday and it was found that it did no help as the groundwater was coming right back into the inlet nozzle location.

Trevor

Trevor Purcell, P.Eng.
 Sr. Site Mechanical Engineer
 Inter Pipeline Ltd. | 55463 Range Road 220 | Fort Saskatchewan, AB T8L-4G1
 O: 1-587-744-0437 | C: 825-440-8105 | E: Trevor.Purcell@interpipeline.com

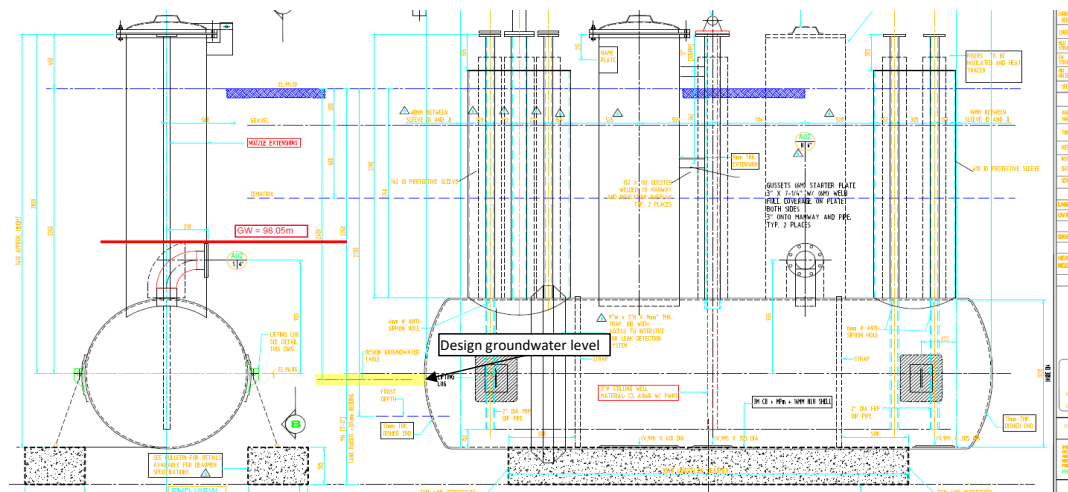
As per the Geotechnical report 19-3500-6, the design elevation of the ground water level is 629.4 m.

It is assumed that there may be up to +/- 0.5 m seasonal fluctuation in the water level.

Therefore, the water level is taken = 630.000m = 98.05m in plant datum elevation.

This level is significantly higher than the 96.8 m ground water elevation used by the Vendor.

Therefore, the installation resistance to buoyancy shall be verified considering the higher water level.



As noted in Thurber's preliminary geotechnical assessment (letter to Williams/Fluor dated July 16, 2013), a preliminary site grade of 631.5 m may be considered based on an average water level of 629.4 m in the main process area as noted in Section 4.3. Water level fluctuations in the order of ± 0.5 m have been recorded at nearby industrial sites, and greater variations may occur during periods of drought or prolonged precipitation events. If a lower risk option is desired that takes this potential variation into account, a site grade elevation of 632.0 m or higher should be selected for the main process areas.

Uplift due to buoyancy effects is checked considering the tank is completely empty.

1.1 Layout of Nozzles:

The top of the tank lies 1.74m below grade elevation.

The high ground water level lies 1.15 m below grade elevation.

Therefore, there is $1.74 - 1.15 = 0.590$ m of submerged Vendor piping nozzles / soil:

Pipe	Nom. Dia	Dia	L _{above water}	L _{below water}	V _{above water}	V _{below water}
N01	2	60.3	1.15	0.590	0.0033	0.0017
V01	4	114.3	1.15	0.590	0.0118	0.0061
N02	2	60.3	1.15	0.590	0.0033	0.0017
M01	24	610	1.15	0.590	0.3356	0.1722
L01	4	114.3	1.15	0.590	0.0118	0.0061
V01	2	60.3	1.15	0.590	0.0033	0.0017
A01	2	60.3	1.15	0.590	0.0033	0.0017
Hatch	32	1,013	1.15	0.590	0.9265	0.4753

: + 2*100mm insulation (assumed insulation thickness)

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TOTALS:	1.30	0.67
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1.2 Tank 10-TA-18005 Buoyancy:

Tank Diameter = 1252 mm
 Length = 4306 mm
 Tank Volume = $\pi/4 * D^2 * L$: ignoring curved ends of tank, conservative
 = 5.30 m³
 Buoyancy Force = (Tank Volume + Nozzle Vol Under Water) * 9.81 kN/m³
 = 58.5 kN

1.3 Resisting Forces to Buoyancy

1.2.1 Self Weight of the Tank

The empty self weight of the tank is taken from vendor drawings:

FABRICATED WEIGHT: 1636 Kg (3608 LBS.) (with LINER)	OPERATING WEIGHT: 8198 Kg
WEIGHT FULL OF WATER: 5924 Kg	ERECTED WT: 1636 Kg

$W_t = 1,636.0 \text{ kg}$
 $= 16.0 \text{ kN}$

1.2.2 Concrete Deadman Block

The tank has supplemental concrete deadman anchor blocks, as supplied by Highland Tank. Catalogue attached.

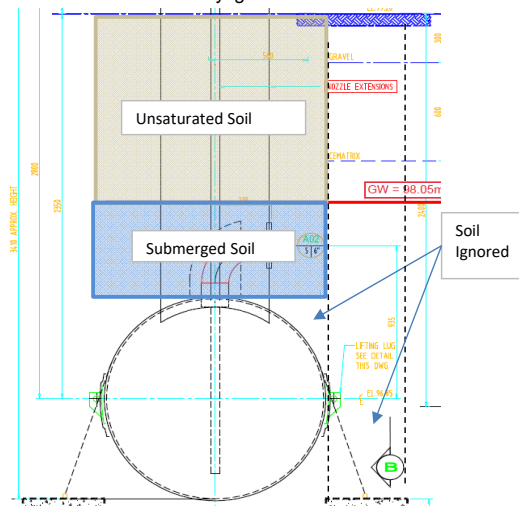
No. of Blocks = 2
 height, h = 305 mm
 weight, w = 457 mm
 length, l = 3048 mm
 Density, γ_c = 23.5 kN/m³
 Effective unit weight, $\gamma_c - \gamma_w$ = 13.7 kN/m³ $\gamma_c - 9.81 \text{ kN/m}^3$
 $W_c = 11.7 \text{ kN}$: 2 * h * w * l * Effective unit weight.

1.2.3 Weight of Soil Cover

Only the soil cover taken above the tank is considered.

Additional soil cover will be present around the curve of the tank and over the anchor blocks.

This additional cover is conservatively ignored.



SOIL DESCRIPTION	BULK UNIT WEIGHT kN/m ³
Sand Backfill - on site sand (compacted to 95% SPMDD)	20
Clay Backfill - on site clay and clay till (compacted to 95% of SPMDD)	21
Pit Run Gravel Compact (compacted to 95% SPMDD)	22

Note: SPMDD = Standard Proctor Maximum Dry Density

* from Geotech report

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Soil Above Water Table:

height, h	=	1.15 m	
weight, w	=	1.25 m	: tank diameter
length, l	=	4.31 m	: tank length
Vol. Nozzles	=	1.30 m ³	: volume of nozzles above water level
Total Volume Soil	=	4.90 m ³	: h*w*I - Vol. Nozzles
Density, γ_s	=	20.0 kN/m ³	
W_{s1}	=	98.0 kN	

Soil Below Water Table:

height, h	=	0.59 m	
weight, w	=	1.25 m	: tank diameter
length, l	=	4.31 m	: tank length
Vol. Nozzles	=	0.67 m ³	: volume of nozzles below water level
Total Volume Soil	=	2.51 m ³	: h*w*I - Vol. Nozzles
Density, γ_s	=	20.0 kN/m ³	
Effective unit weight, $\gamma_s - \gamma_w$	=	10.2 kN/m ³	$\gamma_s - 9.81 \text{ kN/m}^3$
W_{s1}	=	25.6 kN	

$$\text{Total Resistance Force} = W_t + W_c + W_{s1} + W_{s2} = 151 \text{ kN}$$

$$\text{Factor of Safety} = \text{Resisting Force/Buoyancy Force} = 2.59 \text{ OK.}$$

As per Fluor standard practice, 000.215.1240, a buoyancy factor of safety = 1.2 is used for underground concrete pits. The same Safety Factor can be applied here.

Buoyancy

A minimum factor of safety against flotation of 1.2 will be used for the empty condition.

1.4 Anchor Strap Verification

The tank is strapped down to concrete anchor blocks using catalogue Highland Tank Polyester Hold Down Straps.

Tank Dia.	Turn-Buckle Dia.	Strap Length
3'2"	3/4"	73"
4'0"	3/4"	89"

Maximum force acting on strap connection points =

$$(\text{Tank Buoyancy Force} - \text{Weight of Tank}) / 4 \text{ Connection Points} = 10.62 \text{ kN} \\ 2,393 \text{ lbs}$$



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As per correspondence with Highland Tank, the strap/turnbuckle arrangement is rated for the following:

Coltin,

See below response.

We would offer rated strengths of:

48" Dia Strap Kit: 10,400 Lbs. Each

120" Dia Strap Kit: 30,400 Lbs. Each

In both cases the turnbuckles are the weakest link, so this rating is based on that.

Regards,

Matthew Brown, Inside Sales Representative

mbrown@highlandtank.com | 336-478-3293

Highland Tank

[2700 Patterson Street, Greensboro, NC 27407](#)

Stoystown, PA | Manheim, PA | Watervliet, NY | Greensboro, NC | Friedens, PA | Clarkston, MI | Mancelona, MI

The rating is based off of the turnbuckle strength, which is the weakest link in the anchor system.

As can be seen, the 48" strap is rated for 10,400 lbs each, > 2393 lbs. Therefore, the Strap arrangement is acceptable.

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2.0 Sulfuric Acid Drain Tank 10-TA-18004 Buoyancy Verification

Similar to 10-TA-18005, the water level for 10-TA-18004 is higher than indicated on the Vendor GA drawings.

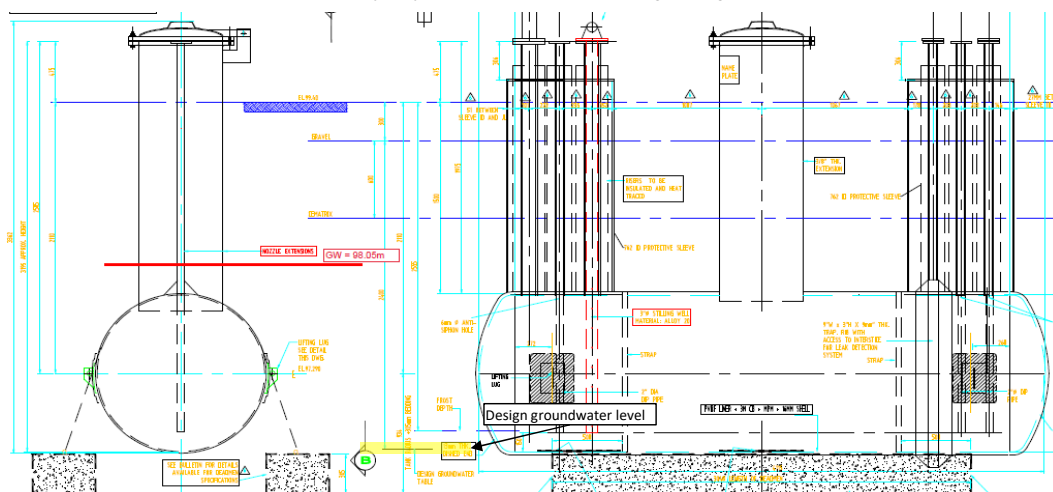
As per the Geotechnical report 19-3500-6, the design elevation of the ground water level is 629.4 m.

It is assumed that there may be up to +/- 0.5 m seasonal fluctuation in the water level.

Therefore, the water level is taken = 630.000m = 98.05m in plant datum elevation.

This level is significantly higher than the 96.7 m ground water elevation used by the Vendor.

Therefore, the installation resistance to buoyancy shall be verified considering the higher water level.



As noted in Thurber's preliminary geotechnical assessment (letter to Williams/Fluor dated July 16, 2013), a preliminary site grade of 631.5 m may be considered based on an average water level of 629.4 m in the main process area as noted in Section 4.3. Water level fluctuations in the order of ± 0.5 m have been recorded at nearby industrial sites, and greater variations may occur during periods of drought or prolonged precipitation events. If a lower risk option is desired that takes this potential variation into account, a site grade elevation of 632.0 m or higher should be selected for the main process areas.

Uplift due to buoyancy effects is checked considering the tank is completely empty.

1.1 Layout of Nozzles:

The top of the tank lies 1.5m below grade elevation.

The high ground water level lies 1.35 m below grade elevation.

Therefore, there is $1.50 - 1.35 = 0.150$ m of submerged Vendor piping nozzles / soil:

Pipe	Nom. Dia	Dia	L _{above water}	L _{below water}	V _{above water}	V _{below water}
V01	4	114.3	1.35	0.150	0.0139	0.0015
A01	2	60.3	1.35	0.150	0.0039	0.0004
L01	4	114.3	1.35	0.150	0.0139	0.0015
M01	24	610	1.35	0.150	0.3940	0.0438
N02	2	60.3	1.35	0.150	0.0039	0.0004
N01	2	60.3	1.35	0.150	0.0039	0.0004
W01	2	60.3	1.35	0.150	0.0039	0.0004
TOTALS:					0.44	0.05

1.2 Tank 10-TA-18004 Buoyancy:

$$\begin{aligned}
 \text{Tank Diameter} &= 1252 \text{ mm} \\
 \text{Length} &= 4115 \text{ mm} \\
 \text{Tank Volume} &= \pi/4 * D^2 * L && \text{: ignoring curved ends of tank, conservative} \\
 &= 5.07 \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Buoyancy Force} &= (\text{Tank Volume} + \text{Nozzle Vol Under Water}) * 9.81 \text{ kN/m}^3 \\
 &= 50.2 \text{ kN}
 \end{aligned}$$

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1.3 Resisting Forces to Buoyancy

1.2.1 Self Weight of the Tank

The empty self weight of the tank is taken from vendor drawings:

FABRICATED WEIGHT: 1630Kg (3600 LBS.) (with LINER)	OPERATING WEIGHT: 9764 Kg
WEIGHT FULL OF WATER: 5916 Kg	ERECTED WT: 1636 Kg

$$\begin{aligned} W_t &= 1,636.0 \text{ kg} \\ &= 16.0 \text{ kN} \end{aligned}$$

1.2.2 Concrete Deadman Block

The tank has supplemental concrete deadman anchor blocks, as supplied by Highland Tank. Catalogue attached.

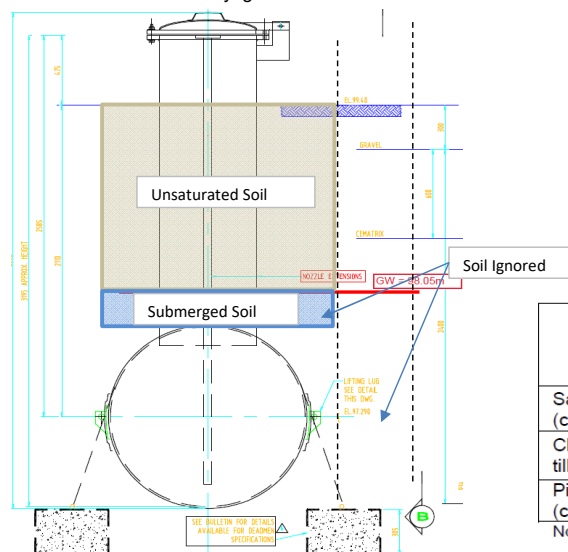
No. of Blocks	=	2	
height, h	=	305 mm	
weight, w	=	457 mm	
length, l	=	3048 mm	
Density, γ_c	=	23.5 kN/m ³	
Effective unit weight, $\gamma_c - \gamma_w$	=	13.7 kN/m ³	$\gamma_c - 9.81 \text{ kN/m}^3$
W_c	=	11.7 kN	: $2 * h * w * l * \text{Effective unit weight.}$

1.2.3 Weight of Soil Cover

Only the soil cover taken above the tank is considered.

Additional soil cover will be present around the curve of the tank and over the anchor blocks.

This additional cover is conservatively ignored.



SOIL DESCRIPTION	BULK UNIT WEIGHT kN/m ³
Sand Backfill - on site sand (compacted to 95% SPMDD)	20
Clay Backfill - on site clay and clay till (compacted to 95% of SPMDD)	21
Pit Run Gravel Compact (compacted to 95% SPMDD)	22

Note: SPMDD = Standard Proctor Maximum Dry Density

* from Geotech report

Soil Above Water Table:

height, h	=	1.35 m	
weight, w	=	1.25 m	: tank diameter
length, l	=	4.12 m	: tank length
Vol. Nozzles	=	0.44 m ³	: volume of nozzles above water level
Total Volume Soil	=	6.52 m ³	: h*w*l - Vol. Nozzles
Density, γ_s	=	20.0 kN/m ³	



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$$W_{s1} = 130.4 \text{ kN}$$

Soil Below Water Table:

height, h	=	0.15 m	
weight, w	=	1.25 m	: tank diameter
length, l	=	4.12 m	: tank length
Vol. Nozzles	=	0.05 m ³	: volume of nozzles below water level
Total Volume Soil	=	0.72 m ³	: h*w*l - Vol. Nozzles
Density, γ_s	=	20.0 kN/m ³	
Effective unit weight, $\gamma_s - \gamma_w$	=	10.2 kN/m ³	$\gamma_s - 9.81 \text{ kN/m}^3$
W_{s1}	=	7.4 kN	

$$\text{Total Resistance Force} = W_t + W_c + W_{s1} + W_{s2} = 165 \text{ kN}$$

$$\text{Factor of Safety} = \text{Resisting Force/Buoyancy Force} = 3.30 \text{ OK.}$$

As per Fluor standard practice, 000.215.1240, a buoyancy factor of safety = 1.2 is used for underground concrete pits. The same Safety Factor can be applied here.

Buoyancy

A minimum factor of safety against flotation of 1.2 will be used for the empty condition.

1.4 Anchor Strap Verification

The tank is strapped down to concrete anchor blocks using catalogue Highland Tank Polyester Hold Down Straps.

Tank Dia.	Turn-Buckle Dia.	Strap Length
3'2"	3/4"	73"
4'0"	3/4"	89"

Maximum force acting on strap connection points =

$$(\text{Tank Buoyancy Force} - \text{Weight of Tank}) / 4 \text{ Connection Points} = \begin{matrix} 8.53 \text{ kN} \\ 1,922 \text{ lbs} \end{matrix}$$

As per correspondence with Highland Tank, the strap/turnbuckle arrangement is rated for the following:

Coltin,

See below response.

We would offer rated strengths of:

48" Dia Strap Kit: 10,400 Lbs. Each

120" Dia Strap Kit: 30,400 Lbs. Each

In both cases the turnbuckles are the weakest link, so this rating is based on that.

Regards,

Matthew Brown, Inside Sales Representative

mbrown@highlandtank.com | 336-478-3293

Highland Tank

2700 Patterson Street, Greensboro, NC 27407

Stoystown, PA | Manheim, PA | Watervliet, NY | Greensboro, NC | Friedens, PA | Clarkston, MI | Mancelona, MI

The rating is based off of the turnbuckle strength, which is the weakest link in the anchor system.

As can be seen, the 48" strap is rated for 10,400 lbs each, > 1922 lbs. Therefore, the Strap arrangement is acceptable.

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3.0 Glycol Drain Tank 10-TA-47001 Buoyancy Verification

Similar to 10-TA-18005, the water level for 10-TA-47001 is higher than indicated on the Vendor GA drawings.

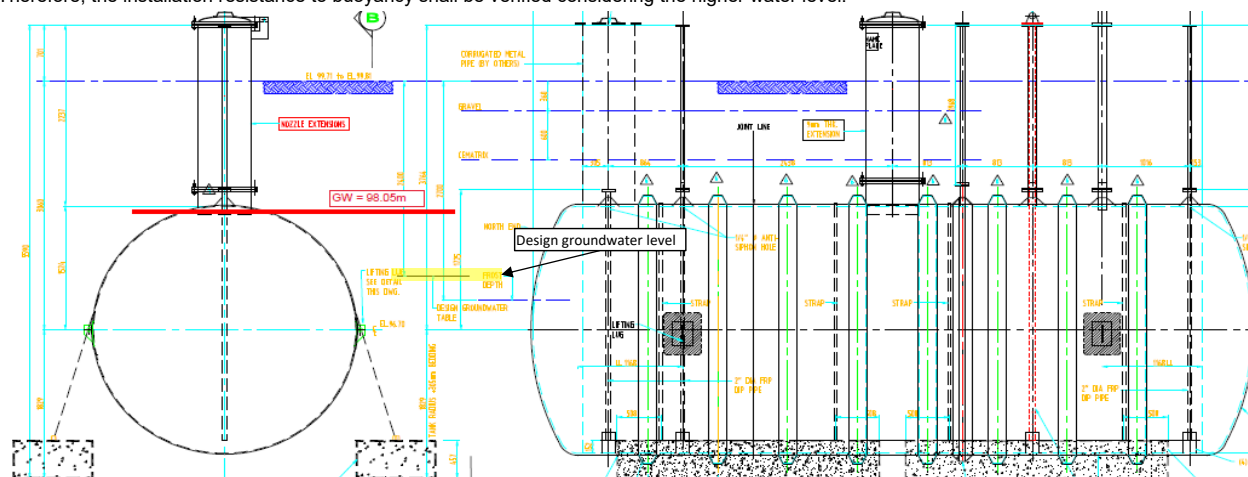
As per the Geotechnical report 19-3500-6, the design elevation of the ground water level is 629.4 m.

It is assumed that there may be up to +/- 0.5 m seasonal fluctuation in the water level.

Therefore, the water level is taken = 630.000m = 98.05m in plant datum elevation.

This level is higher than the 97.31 m ground water elevation used by the Vendor.

Therefore, the installation resistance to buoyancy shall be verified considering the higher water level.



As noted in Thurber's preliminary geotechnical assessment (letter to Williams/Fluor dated July 16, 2013), a preliminary site grade of 631.5 m may be considered based on an average water level of 629.4 m in the main process area as noted in Section 4.3. Water level fluctuations in the order of ± 0.5 m have been recorded at nearby industrial sites, and greater variations may occur during periods of drought or prolonged precipitation events. If a lower risk option is desired that takes this potential variation into account, a site grade elevation of 632.0 m or higher should be selected for the main process areas.

The GW elevation is just below the top of tank. For ease of calculation, this level is conservatively taken equal to top of tank elevation.

Uplift due to buoyancy effects is checked considering the tank is completely empty.

1.1 Layout of Nozzles:

As per above, the GW is taken equal to top of tank elevation.

The top of tank elevation is 1.536 m below finish grade elevation.

Therefore, all of the nozzle piping and soil cover is above the water level.

Pipe	Nom. Dia	Dia	L _{above water}	L _{below water}	V _{above water}	V _{below water}
A01	2	60.3	1.54	0.000	0.004	0.0000
N01	2	60.3	1.54	0.000	0.004	0.0000
M01	24	610	1.54	0.000	0.448	0.0000
N02	2	60.3	1.54	0.000	0.004	0.0000
L01	4	114.3	1.54	0.000	0.016	0.0000
V01	4	114.3	1.54	0.000	0.016	0.0000
W01	2	60.3	1.54	0.000	0.004	0.0000
Hatch	48	1,419	1.54	0.000	2.43	0.0000
TOTALS:					2.93	0.00

: + 2*100mm insulation (assumed insulation thickness)

1.2 Tank 10-TA-47001 Buoyancy:

Tank Diameter = 3048 mm

Length = 8357 mm

Tank Volume = $\pi/4 * D^2 * L$

: ignoring curved ends of tank, conservative

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	=	60.98 m ³
Buoyancy Force	=	(Tank Volume + Nozzle Vol Under Water) * 9.81 kN/m ³
	=	598.2 kN

1.3 Resisting Forces to Buoyancy

1.2.1 Self Weight of the Tank

The empty self weight of the tank is taken from vendor drawings:

FABRICATED WEIGHT:	6350 kg.	OPERATING WEIGHT:	61,001 Kg
WEIGHT FULL OF WATER:	56,409 Kg	ERECTED WT:	6350 Kg

$$\begin{aligned} W_t &= 6,358.0 \text{ kg} \\ &= 62.4 \text{ kN} \end{aligned}$$

1.2.2 Concrete Deadman Block

The tank has supplemental concrete deadman anchor blocks, as supplied by Highland Tank. Catalogue attached.

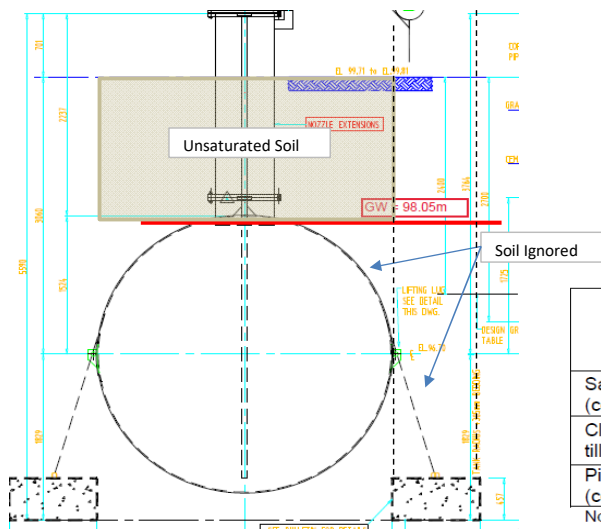
No. of Blocks	=	4	
height, h	=	457 mm	
weight, w	=	914 mm	
length, l	=	3048 mm	
Density, γ_c	=	23.5 kN/m ³	
Effective unit weight, $\gamma_c - \gamma_w$	=	13.7 kN/m ³	$\gamma_c - 9.81 \text{ kN/m}^3$
W_c	=	69.9 kN	: 4 * h * w * l * Effective unit weight.

1.2.3 Weight of Soil Cover

Only the soil cover taken above the tank is considered.

Additional soil cover will be present around the curve of the tank and over the anchor blocks.

This additional cover is conservatively ignored.



SOIL DESCRIPTION	BULK UNIT WEIGHT kN/m ³
Sand Backfill - on site sand (compacted to 95% SPMDD)	20
Clay Backfill - on site clay and clay till (compacted to 95% of SPMDD)	21
Pit Run Gravel Compact (compacted to 95% SPMDD)	22

Note: SPMDD = Standard Proctor Maximum Dry Density

* from Geotech report

Soil Above Water Table:

height, h	=	1.54 m	
weight, w	=	3.05 m	: tank diameter
length, l	=	8.36 m	: tank length

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Vol. Nozzles = 2.93 m³ : volume of nozzles above water level
 Total Volume Soil = 36.20 m³ : h*w*I - Vol. Nozzles
 Density, γ_s = 20.0 kN/m³
 W_{s1} = 724.0 kN

Soil Below Water Table:

height, h = 0.00 m
 weight, w = 3.05 m : tank diameter
 length, l = 8.36 m : tank length
 Vol. Nozzles = 0.00 m³ : volume of nozzles below water level
 Total Volume Soil = 0.00 m³ : h*w*I - Vol. Nozzles
 Density, γ_s = 20.0 kN/m³
 Effective unit weight, $\gamma_s - \gamma_w$ = 10.2 kN/m³ $\gamma_s - 9.81$ kN/m³
 W_{s1} = 0.0 kN

Total Resistance Force = $W_l + W_c + W_{s1} + W_{s2}$ = 856 kN

Factor of Safety = Resisting Force/Buoyancy Force = 1.43 OK.

As per Fluor standard practice, 000.215.1240, a buoyancy factor of safety = 1.2 is used for underground concrete pits. The same Safety Factor can be applied here.

Buoyancy

A minimum factor of safety against flotation of 1.2 will be used for the empty condition.

As noted above: Only the soil cover taken above the tank is considered.
 Additional soil cover will be present around the curve of the tank and over the anchor blocks.
 This additional cover is conservatively ignored.
 The projected volume the tank is used when calculating buoyancy force (ignore reduction of curved ends)

Therefore, this Factor of Safety is conservative.

1.4 Anchor Strap Verification

The tank is strapped down to concrete anchor blocks using catalogue Highland Tank Polyester Hold Down Straps.

Tank Dia.	Turn-Buckle Dia.	Strap Length
3'2"	3/4"	73"
4'0"	3/4"	89"
5'4"	1-1/4"	124"
6'0"	1-1/4"	137"
7'0"	1-1/4"	172"
8'0"	1-1/4"	194"
10'0"	1-1/4"	242"

Maximum force acting on strap connection points =

(Tank Buoyancy Force - Weight of Tank) / 8 Connection Points = 66.98 kN
 15,089 lbs



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As per correspondence with Highland Tank, the strap/turnbuckle arrangement is rated for the following:

Coltin,

See below response.

We would offer rated strengths of:

48" Dia Strap Kit: 10,400 Lbs. Each

120" Dia Strap Kit: 30,400 Lbs. Each

In both cases the turnbuckles are the weakest link, so this rating is based on that.

Regards,

Matthew Brown, Inside Sales Representative
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Stoystown, PA | Manheim, PA | Watervliet, NY | Greensboro, NC | Friedens, PA | Clarkston, MI | Mancelona, MI

The rating is based off of the turnbuckle strength, which is the weakest link in the anchor system.

As can be seen, the 120" strap is rated for 30,400 lbs each, > 15089 lbs. Therefore, the Strap arrangement is acceptable.

Coltin Walsh

From: Matthew Brown <mbrown@highlandtank.com>
Sent: Tuesday, October 12, 2021 8:17 AM
To: Coltin Walsh
Subject: [External] RE: WEBCHAT Polyester Hold Down Strap Question.

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Coltin,

See below response.

We would offer rated strengths of:

48" Dia Strap Kit: 10,400 Lbs. Each
120" Dia Strap Kit: 30,400 Lbs. Each

In both cases the turnbuckles are the weakest link, so this rating is based on that.

Regards,

Matthew Brown, Inside Sales Representative
mbrown@highlandtank.com | 336-478-3293

Highland Tank

2700 Patterson Street, Greensboro, NC 27407
Stoystown, PA | Manheim, PA | Watervliet, NY | Greensboro, NC | Friedens, PA | Clarkston, MI |
Mancelona, MI

From: Coltin Walsh <Coltin.Walsh@fluor.com>
Sent: Tuesday, October 12, 2021 9:42 AM
To: Matthew Brown <mbrown@highlandtank.com>
Subject: RE: WEBCHAT Polyester Hold Down Strap Question.

Good Morning Matthew,

We have two tanks on site – one is a 4'-0" Diameter tank, and the second is a 10'-0" diameter tank:

Tank Diameter	Turn-Buckle Diameter	Strap Length
3'2"	3/4"	73"
4'0"	3/4"	89"
5'4"	1-1/4"	124"
6'0"	1-1/4"	137"
7'0"	1-1/4"	172"
8'0"	1-1/4"	194"
10'0"	1-1/4"	242"
10'6"	1-1/4"	242"

Thanks!

Coltin Walsh, P.Eng (AB) | **FLUOR CANADA LTD** | Civil Structural Architectural Engineer | coltin.walsh@fluor.com
 IODC 40.4259 | C +1.403.537.4259
www.fluor.com | [Facebook](#) | [Twitter](#) | [LinkedIn](#) | [YouTube](#)

From: Matthew Brown <mbrown@highlandtank.com>
Sent: Friday, October 8, 2021 6:39 AM
To: Coltin Walsh <Coltin.Walsh@fluor.com>
Subject: [External] WEBCHAT Polyester Hold Down Strap Question.

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Coltin,

What is the diameter of the tank/strap length you are working on?

Regards,

Matthew Brown, Inside Sales Representative
mbrown@highlandtank.com | 336-478-3293
Highland Tank
 2700 Patterson Street, Greensboro, NC 27407
 Stoystown, PA | Manheim, PA | Watervliet, NY | Greensboro, NC | Friedens, PA | Clarkston, MI |
 Mancelona, MI

Highland Tank Confidentiality Notice:

Highland Polyester Hold Down Straps

The World's Easiest Hold Down Strap System!



Highland Tank®



Highland Tank

Polyester Hold-down Straps

The latest tank innovation from the nation's leading manufacturer - Highland Tank. Polyester Straps are designed to help secure tanks underground in areas having ground water.

Strap liners are not needed! If installed correctly, there are not any metallic components in the strap system that can make contact with the tank.

Hold-down Straps are made from 100% polyester webbing. Highland's Polyester Straps are fabricated with a large reinforced loop at each end for securing your underground cylindrical storage vessel. Included with each strap is a length of 1/2" diameter wire rope and three cable clamps for attaching the strap to one side of your concrete anchoring system. A turnbuckle is used for attaching the strap to the anchoring on the other side of the tank.

Tank installation instructions should be referenced for specific strap requirements.

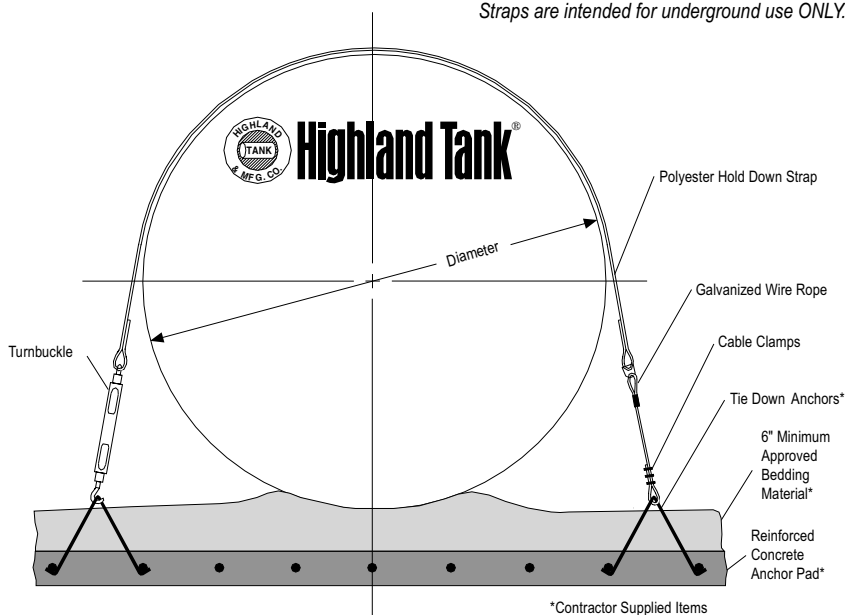
Installations Instructions

- Open turnbuckle the whole way.
- Lay polyester strap over tank.
- Attach one end of turnbuckle to loop of polyester strap.
- Attach other end of turnbuckle through tie-down anchor (by contractor) on one side of tank.
- Go to other side of tank.
- Wrap 1/2" wire rope attached to the other end of the polyester strap through the tie-down anchor (by contractor).
- Pull rope tight. Insert thimble at wire rope and tie-down juncture. Secure wire rope with (3) supplied cable clamps.
- Go back to turnbuckle side of vessel.
- Tighten turnbuckle to secure tank.
- After taking up slack in the polyester strap, tighten turnbuckle an additional 2".

These straps are designed for and must be used in a system that has adequately sized concrete anchoring to counteract the tank's natural buoyant forces. Polyester Straps are intended for underground use ONLY.

Tank Dia.	Turn-Buckle Dia.	Strap Length
3'2"	3/4"	73"
4'0"	3/4"	89"
5'4"	1-1/4"	124"
6'0"	1-1/4"	137"
7'0"	1-1/4"	172"
8'0"	1-1/4"	194"
10'0"	1-1/4"	242"

Hold-Down Straps for tanks larger than 10'0" diameter are available. Please contact Highland at one of the locations listed below.



Please visit us at www.highlandtank.com

One Highland Road
Stoystown, PA 15563
814-893-5701
FAX 893-6126

4535 Elizabethtown Road
Manheim, PA 17545
717-664-0600
FAX 664-0617

958 19th Street
Watervliet, NY 12189
518-273-0801
FAX 273-1365

2700 Patterson Street
Greensboro, NC 27407
336-218-0801
FAX 218-1292

2225 Chestnut Street
Lebanon, PA 17042
717-664-0602
FAX 664-0631

1510 Stoystown Road
Friedens, PA 15541
814-443-6800
FAX 444-8662

Concrete Deadmen Anchors

Company's Contract/Purchase Order Number- A6KV-4-0208-01
Document Title- Concrete Deadmen Anchors
Document Revision-0
Equipment Tag(s)-10-TA-18004, 10-TA-18005 & 10-TA-47001
Electronic document file name - A6KV-4-0208-01-00048

 inter pipeline **FLUOR**, Client PO No: 7428
Project Number: A6KV/1202 Rec'd Date: Jul 19, 2017
Document No. / Rev: A6KV-4-0208-01-00048_R1
Equipment Tags (if applicable):

10-TA-18004; 10-TA-18005; 10-TA-47001

Economical and Effective Anchoring

Highland Tank



D - REVIEWED FOR INFORMATION ONLY **FLUOR**
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and/or Purchase Order.
By: Mayank, MIT38381, 7/24/2017

Concrete Deadmen Anchors

Highland adds to our extensive product line by delivering pre-fabricated Concrete Deadmen Anchors (CDA) along with our high-quality steel storage and wastewater treatment tanks.

Deadmen Anchors are beams of reinforced concrete placed alongside the tank in the bottom of the excavation. The weight of the backfill on the Deadmen Anchors provides additional resistance to buoyancy forces acting on the tank.

Properly installed Deadmen Anchors, when used exclusively with Highland's Polyester Hold-Down Straps and supplied hardware, prevent tank flotation and cost less than a concrete bottom hold-down pad.

Best of all, there is no delivery charge for Concrete Deadmen Anchors when they are shipped on the same truck along with a Highland tank product!

Highland's pre-engineered concrete deadmen anchors work with our standard horizontal underground UL-58 tank products ranging from 500 to 50,000 gallons.

A limited number of Deadmen Anchors require the installation of an 8" thick grade-level reinforced concrete pad (as recommended by PEI/RP 100-05).

Deadmen Anchor recommendations assume that tanks are buried at least 36" deep and have at least one Tank Top Sump installed. They are designed to work with Highland's Polyester Hold-Down Straps. Other methods are readily available and should be specified by a licensed professional engineer.

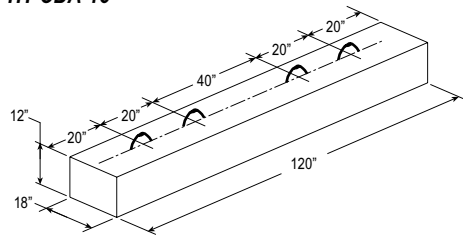
If your requirements are different, please forward details of the application to our engineering department who will perform an evaluation of your intended installation.

Standard Features

- Designed to work with standard underground storage tanks from 500 to 50,000 gallons.
- Utilize a 4,000 psi concrete mix reinforced with #4 rebar.
- Dimensions:
 - CDA-15: 120" L, 18" W, 12" H
 - CDA-45: 120" L, 36" W, 18" H
- Volume:
 - CDA-15: 15 cubic feet (approx.)
 - CDA-45: 45 cubic feet (approx.)
- Weight Each:
 - CDA-15: 2,200 pounds (approx.)
 - CDA-45: 6,500 pounds (approx.)

Concrete Deadmen Anchors can also be used with Highland's other standard underground installed products such as our Oil/Water Separators, Oil/Sand Interceptors, Solids Stormwater Interceptors, Passive Grease Interceptors and HighDRO™ Water Tanks.

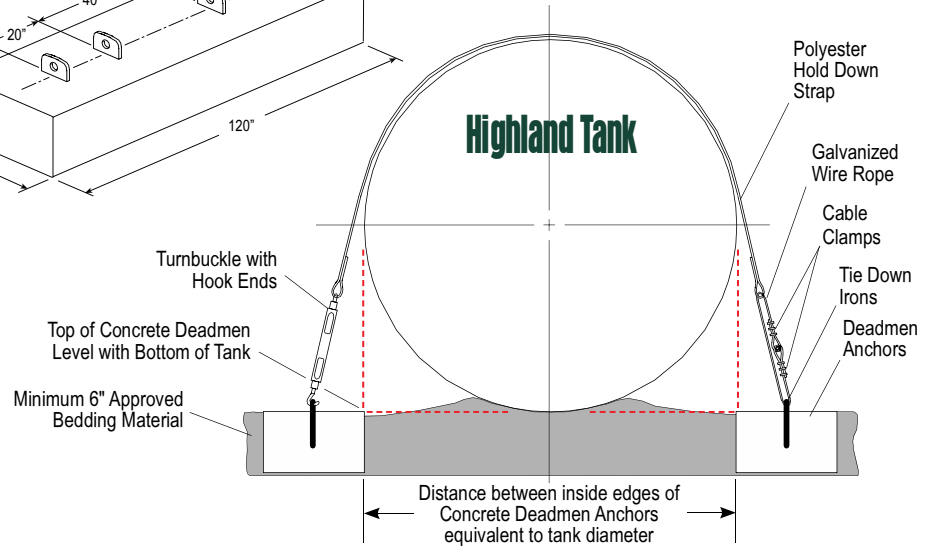
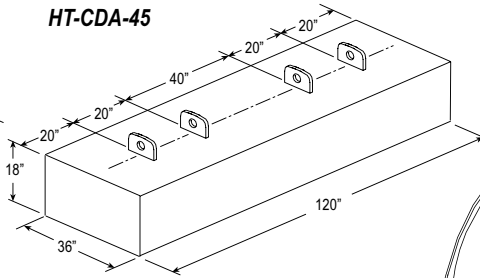
General Arrangement HT-CDA-15



The diagram at right depicts a typical installation that employs Highland's Polyester Hold Down Straps and Concrete Deadmen Anchors. Steel cable and cable clamps are shown to illustrate one method of connecting the straps to the concrete deadman anchors. Other methods are readily available and should be specified by a licensed professional engineer.

Deadmen Anchors must be installed on both sides and along the full length of the tank. They must be positioned so the edges of the deadmen are aligned outside the tank "shadow;" outside the tank diameter and below the tank bottom.

HT-CDA-45



Tank Volume Gallons	Tank Dimensions Diameter	Length	Required Deadmen	Required Straps
500	4'-0"	5'-5"	2	2
1,000	4'-0"	10'-9"	2	2
1,000	5'-4"	6'-0"	2	2
1,500	5'-4"	9'-0"	2	2
2,000	5'-4"	12'-0"	2	2
2,500	5'-4"	15'-0"	2	2
3,000	5'-4"	18'-0"	2	2
4,000	5'-4"	24'-0"	4	4
4,000	6'-0"	19'-0"	4	4
4,000	8'-0"	10'-8"	2	2
5,000	6'-0"	23'-10"	4	4
5,000	8'-0"	13'-4"	2	2
6,000	6'-0"	28'-8"	6	6
6,000	8'-0"	16'-0"	2	2
8,000	8'-0"	21'-4"	4	4
*8,000	10'-0"	14'-0"	2	2
10,000	8'-0"	26'-8"	6	6
*10,000	10'-0"	17'-0"	4	4
12,000	8'-0"	32'-0"	6	6
*12,000	10'-0"	20'-6"	4	4
15,000	8'-0"	40'-0"	8	8
*15,000	10'-0"	25'-6"	4	4
*20,000	10'-0"	34'-0"	6	6
*20,000	10'-6"	31'-0"	6	6
*25,000	10'-6"	38'-9"	8	8
*30,000	10'-6"	46'-0"	10	10
†*40,000	12'-0"	47'-6"	10	10
†*50,000	12'-0"	59'-6"	12	12

*CDA-45 required. All calculations assume tanks are buried at least 36" deep, have at least one sump attached and will be installed exclusively with Highland's Polyester Hold-Down Straps. Contact Highland Tank for custom tank installation recommendations. Accuracy and/or correctness of tank installation is the responsibility of the Engineer of Record and the Installer.

†These sizes currently must employ one of Highland's steel hold-down strap systems.



Please visit us at www.highlandtank.com

One Highland Road
Stoystown, PA 15563
814-893-5701
FAX 893-6126

4535 Elizabethtown Road
Manheim, PA 17545
717-664-0600
FAX 664-0617

958 19th Street
Watervliet, NY 12189
518-273-0801
FAX 273-1365

2700 Patterson Street
Greensboro, NC 27407
336-218-0801
FAX 218-1292

2225 Chestnut Street
Lebanon, PA 17042
717-664-0602
FAX 664-0631

1510 Stoystown Road
Friedens, PA 15541
814-443-6800
FAX 444-8662

DESIGN DATA	
Design Basis: ULC S615	
TANK CONTENTS:	9.99% SULFURIC ACID
SPECIFIC GRAVITY:	1.7
DESIGN PRESSURE (INT/EXT):	65.24 / 51.29 kPag (SOIL LOAD)
WORKING PRESSURE:	ATMOSPHERIC
DESIGN TEMPERATURE:	-29° to 65° C
FABRICATED WEIGHT: 163Kg (358 LBS) (with LINER)	OPERATING WEIGHT: 9764 Kg
WEIGHT FULL OF WATER: 59% Kg	ERECTED WT: 1636 Kg
RESIN TEMPERATURE RATING:	65.5
MAXIMUM FILLING RATE:	611 kg/hr NITROGEN
MAXIMUM EMPTYING RATE:	611 kg/hr NITROGEN
HYDROTEST PRESSURE:	-35 kPag
WIND / SEISMIC DATA:	NOT APPLICABLE
INSULATION/ HEAT TRACING:	REFER TO DWG PDH-E33-TA18004-R1
PAINTING:	NOT REQUIRED - RESIN IS PIGMENTED
TANK WORKING CAPACITY:	4 CUBIC/M
SHIPPING DIMENSIONS:	1524mm WIDE x 3600mm HIGH x 4300mm LG. APPROX.

PRECISIONEERING

303 NANTUCKET BLVD. SCARBOROUGH, ONTARIO

SHOP ORDER

TH16593-2

DATE MFD.

2018

PLANT NAME

PROPANE DEHYDROGENATION PLANT

PURCHASE ORDER

A6KV-4-0208-01

EQUIP. NO.

10-TA-18004

DESCRIPTION

1220 x 3594 STRAIGHT SIDE TANK

SERVICE DESCRIPTION

SULFURIC ACID DRAIN TANK

DESIGN PRESS.

65.24 / 51.27 kPag

DESIGN TEMP.

65° C

WORKG. PRESS.

ATMOSPHERIC

WORKG. TEMP.

5-36° C

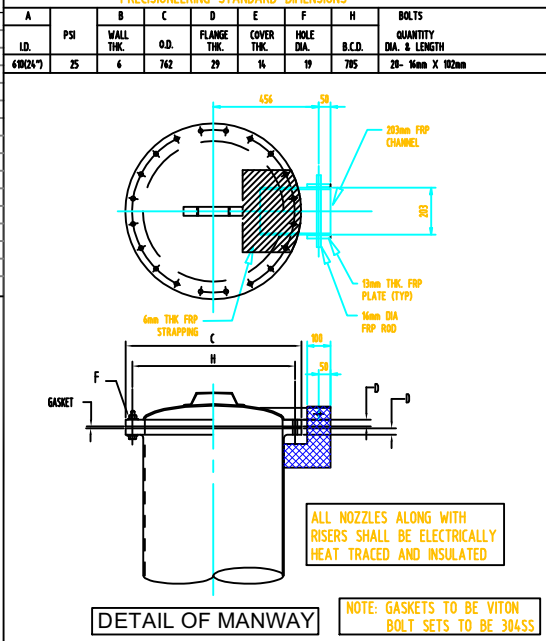
MATERIAL

PVDF LINER + DERAKANE 470-300/411-350

TANK NAMEPLATE

METRIC DIMENSIONS

PRECISIONEERING STANDARD DIMENSIONS



DETAIL OF MANWAY

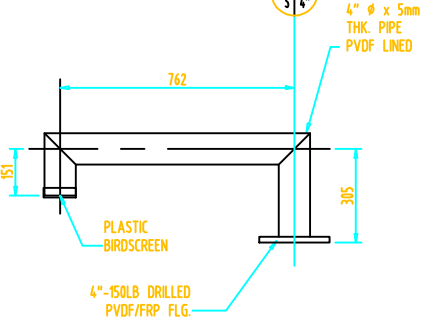
NOTES:

1. MATERIAL - STAINLESS STEEL- 304SS
2. REMOVE ALL BURRS, SHARP EDGES AND ROUND ALL CORNERS

LIFTING LUG DETAIL

Interpipeline FLUOR Client PO No: 66794
PROPANE DEHYDROGENATION PROJECT
Project Number: A6KV1202 Rec'd Date: Jul 16, 2018
Document No. / Rev: A6KV-4-0208-01-00002 R7
Equipment Type (if applicable)

10-TA-18004



SPARES INCLUDED

TAG 10-TA-18005/0001

200% GASKETS (MINIMUM 2 Nos.) FOR MANWAYS, NOZZLES WITH BLIND FLANGES (1.00)

TAG 10-TA-18005/0002

10% BOLTS AND NUTS (MINIMUM 2 Nos.) FOR MANWAYS, NOZZLES WITH BLIND FLANGES (1.00)

NOTE-1

TRACE TEK LEAK DETECTOR SYSTEM SUPPLIED BY PRECISIONEERING c/w SENSOR INTERFACE MODULE WITH RELAY , 120V

FRP TANK NOZZLE LOADS

ALLOWABLE LOAD IN ANY DIRECTION FX,FY or FZ IS 1,112N
ALLOWABLE MOMENT MX MY or MZ IS 166 Nm

NOTE-2

BACKFILL AND COMPACT INSIDE PROTECTIVE SLEEVES ALONG WITH THE REST OF BACKFILLING

AW - PROCEED / WITH COMMENTS **FLUOR**
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and/or Purchase Order.
By: Esther Chan, CHA70201, 7/23/2018

NOZZLE SCHEDULE

	NOZZLE SCHEDULE							SHOP ORDER NO. TH16593-2		DRAWING NO. D-16593-2		REV 6	
MARK	SIZE	DROLLING	RATING	FLANGE THICKNESS	CONN TYPE	NOZ PROJ	STRAP THK.	STRAP WIDTH ON SHELL	GASKET	INTERNAL PIPE	PFG. CODE	DESCRIPTION	I.C. CHECK
AN1	2"	300 LB.	300PSI	1-1/8"	FIG. 6	152	6mm	102	-	YES	HLU	DRAIN SULFURIC ACID INLET w/ DIP PIPE	
HN1	24"	STD	25PSI	1-3/8"	FIG. 6	152	19mm	305	-	-	HLU	MANWAY c/w BLIND & DAVIT	
HN1	2"	300 LB.	300PSI	1-1/8"	FIG. 6	152	6mm	102	-	YES	HLU	UTILITY CONNECTION w/ DIP PIPE w/ BLIND	
HN2	2"	300 LB.	300PSI	1-1/8"	FIG. 5	152	6mm	102	-	-	HLU	LEAK DETECTOR c/w CPVC BLIND FLG. NO PVDF	
V01	4"	150 LB.	150PSI	1-1/4"	FIG. 5	152	6mm	102	-	-	HLU	VENT w/ GOOSENECK AND BROSSCREEN	
WN1	2"	300 LB.	300PSI	1-1/8"	FIG. 5	152	6mm	102	-	YES	HLU	DRAINED SULFURIC ACID OUTLET w/DIP PIPE	
L01	4"	300 LB.	300PSI	1-1/4"	FIG. 5	152	7mm	127	-	YES	HLU	LEVEL TRANSMITTER	

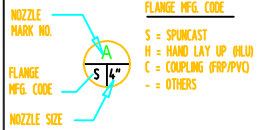
BLIND FLANGES & COVERS

ITEM	SIZE	QTY	DROLLING	RATING	THK	DESCRIPTION	GASKET	BOLTS
1	24"	1	STD	25PSI	STD	TOP MANWAY-H01	1/8" VITON	304SS

MATING FLANGE GASKETS & FASTENERS

ITEM	SIZE	QTY	DROLLING	RATING	DESCRIPTION	GASKET	BOLTS
1	4"	1	150LB	150PSI	NOZZLE- V01	1/8" VITON	304SS
2	4"	1	300LB	300PSI	NOZZLE- L01	1/8" VITON	304SS
3	2"	2	300LB	300PSI	NOZZLE- WN1, NO2	1/8" VITON	304SS

NOTE: UNLESS NOTED OTHERWISE, FLANGE DIMENSIONS SHALL BE AS PER ASME B 16.5



CONSTRUCTION DETAILS

CORROSION LINER REINFORCING	PVDF LINER		
LINER	DERAKANE 470-300	COLOR	CLEAR
HLU STRUCTURAL	DERAKANE 411-350	COLOR	GREY
FR STRUCTURAL	N/A	COLOR	GREY
HOT COAT OUTER SKIN	DERAKANE 411-350	COLOR	GREY
SPECIFICATION	DESIGN AND BUILT TO ULC S615		

QUALITY CONTROL

BARCOL HARDNESS	MINIMUM	TOP HEAD	SHELL	BOTTOM
THICKNESS	TOP HEAD	SHELL	BOTTOM	MANWAY BUILT-UP
INTERIOR CLEAN	WITNESSED BY: DATE:			
INSPECTED BY:	DATE:			
GENERAL ORIENTATION	DETAILED ORIENTATION			

TANK REQUIREMENTS

LAMINATE LEVEL	STD.	LIFTING LUGS	YES
UV PROTECTED	YES	PYRAMID GUSSETS [P]	-
SAVE CUT-OUTS	-		
HEAT TRACING NOZZLES AND RISERS	YES		

TANK HANDLING INSTRUCTIONS

- TO LIFT TANK, USE CRANE AND SPREADER BAR.
- USE ONLY VERTICAL LIFT ON ALL CABLES

DOUBLE WALL FRP TANK

REFERENCE AN-COR JOB NO. 18-2171-D1

CERTIFIED FOR PRODUCTION

BY: DATE:

DO NOT SCALE THIS DRAWING USE DIMENSIONS ONLY

THE INFORMATION AND KNOW HOW ON THIS DRAWING ARE CONFIDENTIAL AND MAY NOT BE USED OR REPRODUCED WITHOUT WRITTEN PERMISSION OF PRECISIONEERING

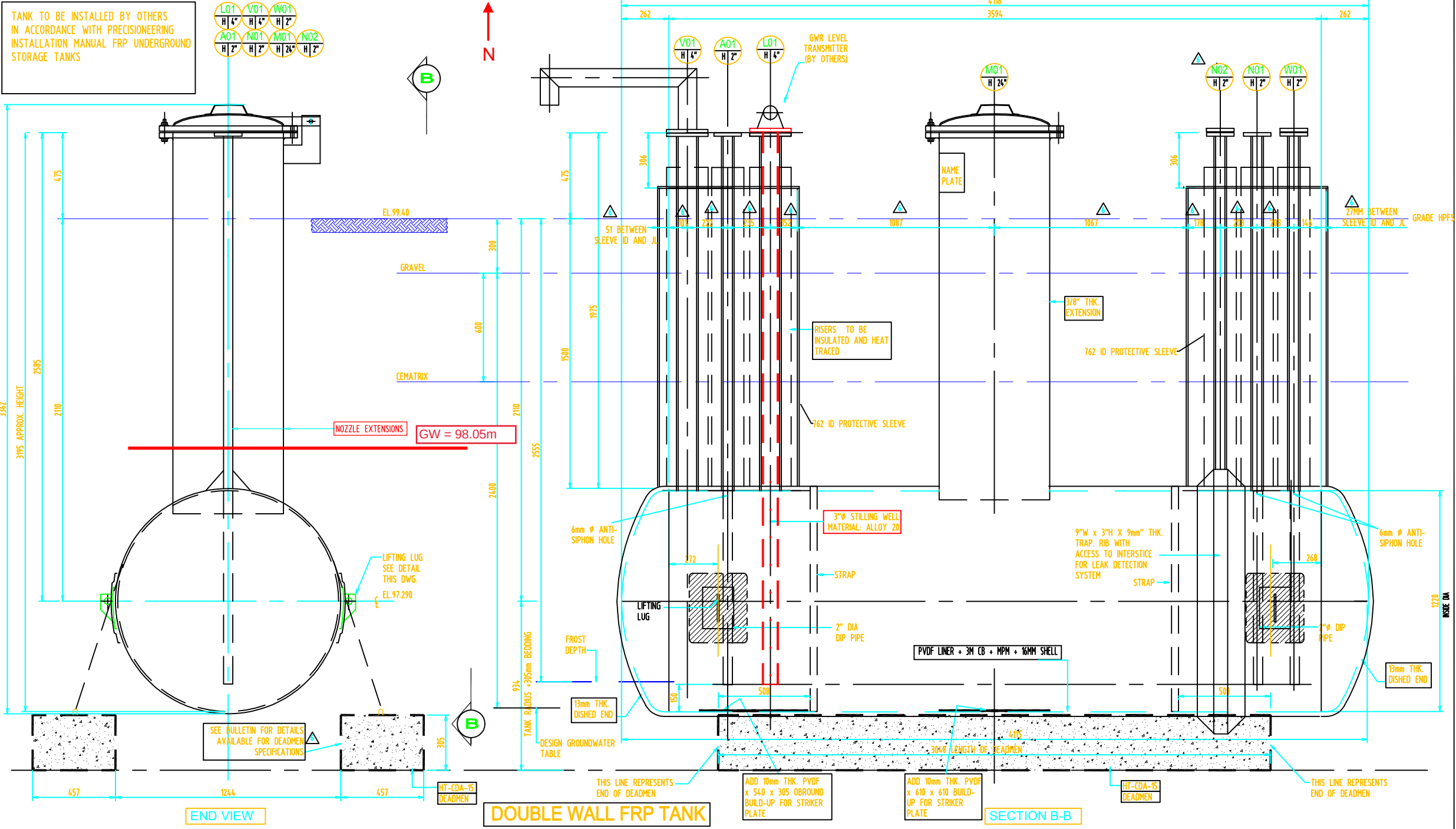
CAD FILE NO: TH16593-2
CUSTOMER REFERENCE DWG: A6KV-4-0208-01-00002

DRAWN: W Inumerable
CHECKED: Brock Humphrey
DATE: MAY 2016
SCALE: 1" = 1'-0"

CUSTOMER: Inter Pipeline Propylene Ltd
PROJECT: Propane Dehydrogenation Project
PROJECT LOCATION: Strathcona County, Alberta, Canada
EQUIPMENT TAG NO.: 10-TA-18004
DOCUMENT TITLE: General Arrangement

PURCHASE ORDER NO.: A6KV-4-0208-01
SHOP ORDER NO.: TH16593-2
DRAWING NO.: D-16593-2
REV: 6

TANK TO BE INSTALLED BY OTHERS IN ACCORDANCE WITH PRECISIONEERING INSTALLATION MANUAL FRP UNDERGROUND STORAGE TANKS



DESIGN DATA	
Design Basis: ULC S615	
TANK CONTENTS:	50% CAUSTIC
SPECIFIC GRAVITY:	1.52
DESIGN PRESSURE (INT.EXT.):	55.38 / 55.80 kPag (SOIL LOAD)
WORKING PRESSURE:	+ 0.62 to -0.36 kPag
DESIGN TEMPERATURE:	-29° to 65° C
FABRICATED WEIGHT: 1634kg (3600 LBS) (with LINER)	OPERATING WEIGHT: 8138 Kg
WEIGHT FULL OF WATER: 5924 Kg	ERECTED WT: 1636 Kg
RESIN TEMPERATURE RATING:	65.5
MAXIMUM FILLING RATE:	611 kg/hr NITROGEN
MAXIMUM EMPTYING RATE:	611 kg/hr NITROGEN
HYDROTEST PRESSURE:	-35 kPag
WIND / SEISMIC DATA:	NOT APPLICABLE
INSULATION/ HEAT TRACING:	REFER TO DWG PDH-E33-TANKWES-01
PAINTING:	NOT REQUIRED , RESIN IS PIGMENTED
TANK WORKING CAPACITY:	4 CUBIC/M
SHIPPING DIMENSIONS:	1524mm WIDE x 3800mm HIGH x 4500mm L.G. APPROX.

PRECISIONEERING

303 NANTUCKET BLVD. SCARBOROUGH, ONTARIO

SHOP ORDER

TH16593-3

DATE MFD.

2018

PLANT NAME

PROPANE DEHYDROGENATION PLANT

PURCHASE ORDER

ABKV-4-0208-01

EQUIP. NO.

10-TA-18005

DESCRIPTION

1220 x 3784 STRAIGHT SIDE TANK

SERVICE DESCRIPTION

CAUSTIC DRAIN TANK

DESIGN PRESS.

55.38 / 55.80 kPag

DESIGN TEMP.

65° C

WORKG. PRESS.

ATMOSPHERIC

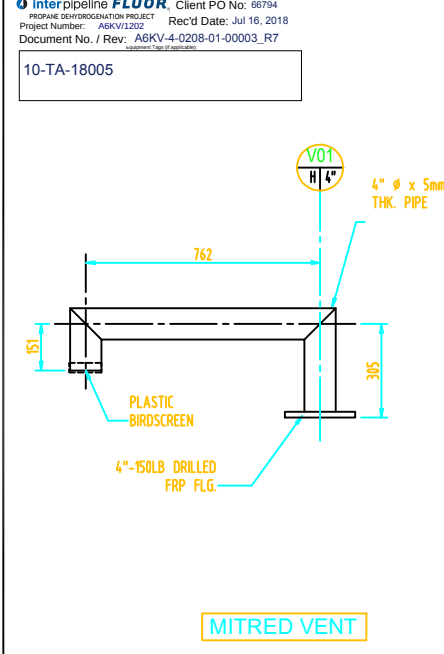
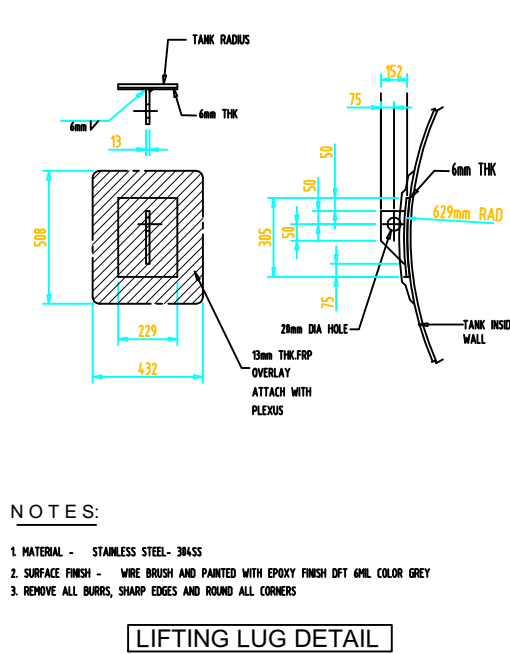
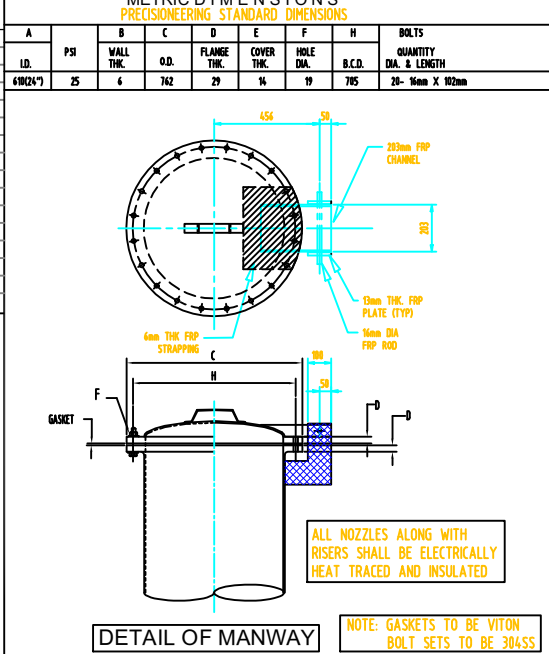
WORKG. TEMP.

5-45° C

MATERIAL

DERAKANE 411-350

TANK NAMEPLATE



SPARES INCLUDED

TAG 10-TA-18005/0001 200% GASKETS (MINIMUM 2 Nos.) FOR MANWAYS, NOZZLES WITH BLIND FLANGES (1.00)

TAG 10-TA-18005/0002 10% BOLTS AND NUTS (MINIMUM 2 Nos.) FOR MANWAYS, NOZZLES WITH BLIND FLANGES (1.00)

NOTE-1

TRACE TEK LEAK DETECTOR SYSTEM SUPPLIED BY PRECISIONEERING c/w SENSOR INTERFACE MODULE WITH RELAY , 120V

FRP TANK NOZZLE LOADS

ALLOWABLE LOAD IN ANY DIRECTION FX,FY or FZ IS 1,112N
ALLOWABLE MOMENT MX MY or MZ IS 166 Nm

NOTE-2

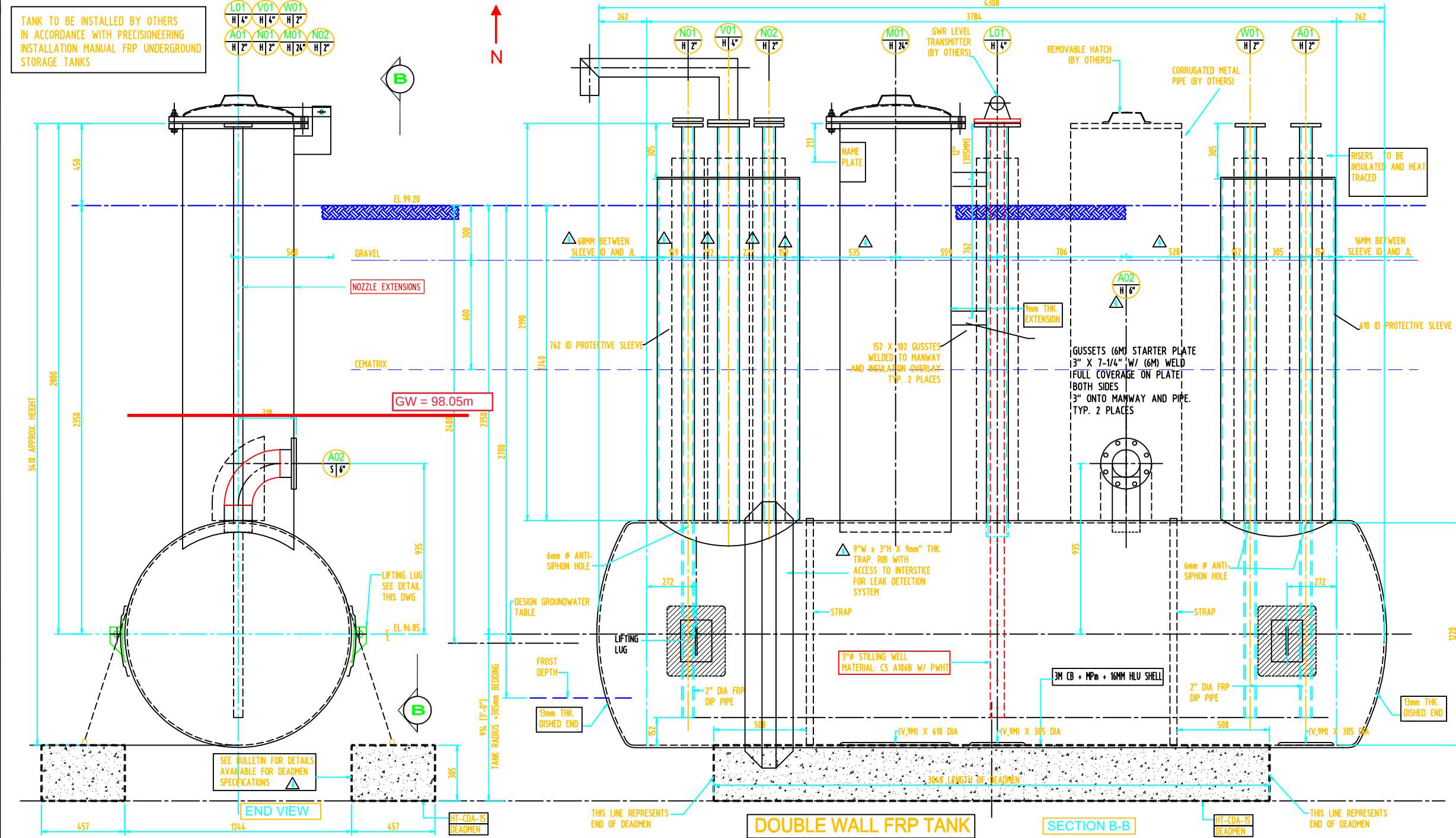
BACKFILL AND COMPACT INSIDE PROTECTIVE SLEEVES ALONG WITH THE REST OF BACKFILLING

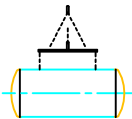
AW - PROCEED / WITH COMMENTS **FLUOR**

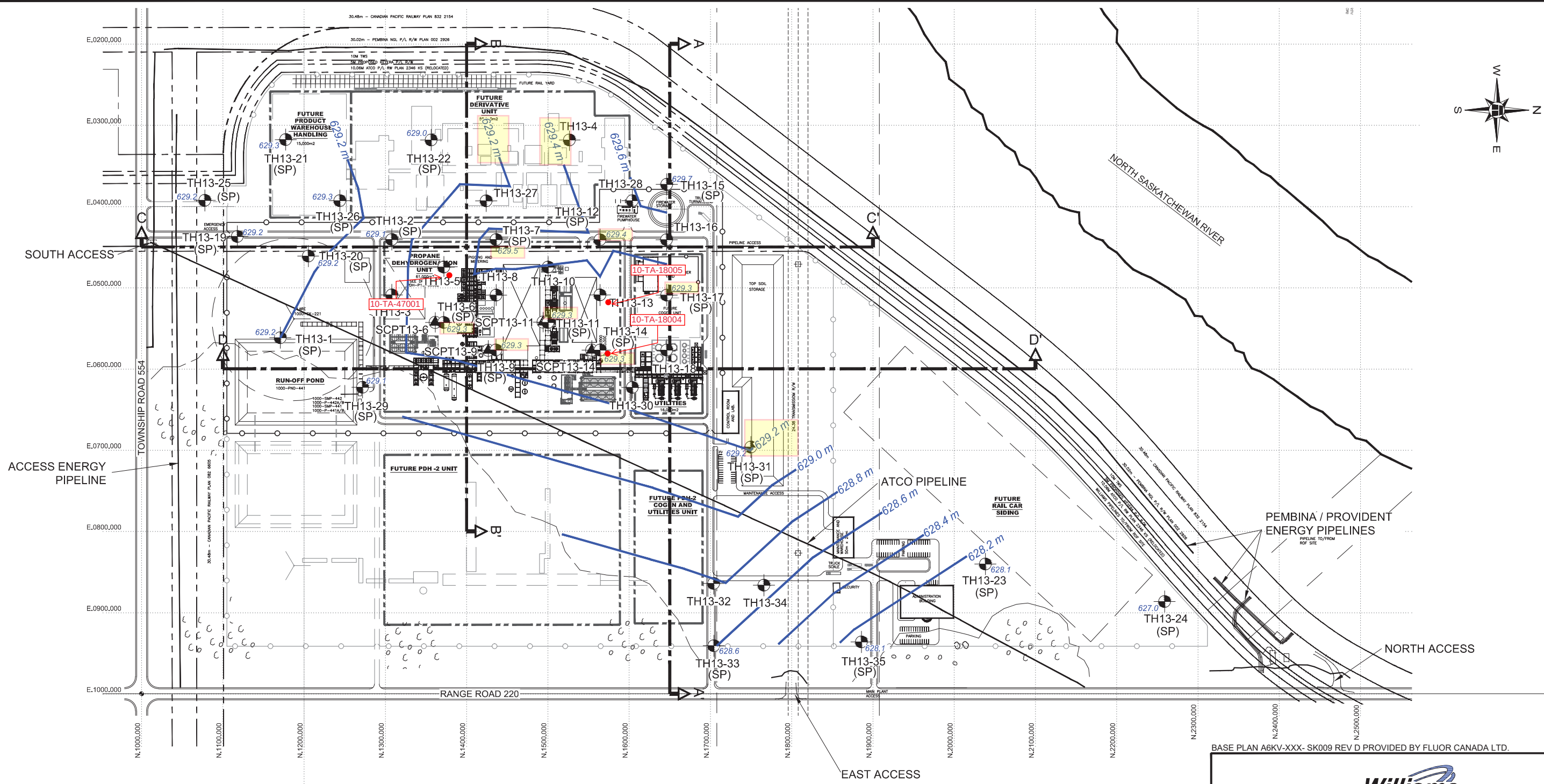
Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and/or Purchase Order.

By: Esther Chan,CHA70201,7/23/2018

NOZZLE SCHEDULE										SHOP ORDER NO. TH16593-3		DRAWING NO. D-16593-3		6
MARK	SIZE	DRAWING	RATING	FLANGE THICKNESS	CONN. TYPE	NOZ. PROJ.	STRAP THK.	STRAP WIDTH ON SHELL	GUSSET	INTERNAL GASK.	HPG. CODE	DESCRIPTION		O.C. CHECK
A01	2"	300 LB.	30MPSI	7/8"	FIG. 6	52	6mm	102	-	YES	HLU	DRAIN 20% CAUSTIC ACID INLET w/ DIP PIPE		
M01	24"	STD.	25PSI	1-1/8"	FIG. 6	52	75mm	385	-	-	HLU	MANWAY clw BLIND & DAVIT		
N01	2"	300 LB.	30MPSI	7/8"	FIG. 6	52	6mm	102	-	YES	HLU	UTILITY CONNECTION w/ DIP PIPE w/BLIND		
N02	2"	300 LB.	30MPSI	7/8"	FIG. 5	52	6mm	102	-	-	HLU	LEAK DETECTOR w/ CPVC BLIND		
V01	4"	150 LB.	15MPSI	1"	FIG. 5	52	6mm	102	-	-	HLU	VENT w/ GOOSENECK AND BROSSCREEN		
W01	2"	300 LB.	30MPSI	7/8"	FIG. 6	52	6mm	102	-	YES	HLU	DRAIN CAUSTIC ACID OUTLET w/ DIP PIPE		
L01	4"	300 LB.	30MPSI	1" (T2)	FIG. 6	52	7mm	127	-	YES	HLU	LEVEL TRANSMITTER		
A02	6"	300 LB.	15MPSI	5/16" (T2)	FIG. 6	52	7mm	127	-	YES	HLU	WASTE WATER DRAIN INLET		
BLIND FLANGES & COVERS														
ITEM	SIZE	QTY	DRAWING	RATING	THK.	DESCRIPTION					GASKET	BOLTS		
1	24"	1	STD.	25PSI	STD.	TOP MANWAY-M01					1/8" VITON	304SS		
2	2"	2	300 LB.	30MPSI	STD.	FOR NOZZLE-N01, N02					1/8" VITON	304SS		
MATING FLANGE GASKETS & FASTENERS														
ITEM	SIZE	QTY	DRAWING	RATING	DESCRIPTION					GASKET	BOLTS			
1	2"	2	300LB.	30MPSI	NOZZLE- NO1, N02					1/8" VITON	304SS			
2	4"	1	150LB.	15MPSI	NOZZLE- V01					1/8" VITON	304SS			
NOTE: UNLESS NOTED OTHERWISE, FLANGE DIMENSIONS SHALL BE AS PER ASME B 16.5														
NOZZLE MARK NO. 1A FLANGE HPG. CODE S/4" NOZZLE SIZE														
NOZZLE TYPE														
FIG. 6 FF. ASTM D2009 FIG. 6														
FIG. 5 FF. ASTM D2009 FIG. 5														
FLANGE HOLES BOLT HOLES STRADDLE TANK MAJOR CENTERLINES UNLESS NOTED OTHERWISE														
RATING FLANGE FIBERGLASS FLANGES ON TANK MUST ALWAYS BE CONNECTED TO FLAT FACE FLANGES ON EXTERNAL PIPING DO NOT CONNECT TO RAISED FACE FLANGES UNLESS SPACER RING & FULL FACE GASKET IS USED														
FLANGE BOLT TORQUE PLEASE FOLLOW VALUES ON ATTACHED DOCUMENT														



SHOP ORDER NO. TH16593-3				CONSTRUCTION DETAILS				LAMINATE SEQUENCE							
CORROSION LINER REINFORCING				2-SYNTHETIC VEIL + 2 MAT				THK. in mm		HLU LAMINATE					
LINER		DERAKANE 411-350		COLOR		CLEAR		5		SS - 3M/MPM/2M. H					
HLU STRUCTURAL		DERAKANE 411-350		COLOR		GREY		6		SS - 3M/MPM/M R M M. H					
FR STRUCTURAL		N/A		COLOR		GREY		7		SS - 3M/MPM/20MR) M. H					
HOT COAT OUTER SKIN		DERAKANE 411-350		COLOR		GREY		9		SS - 3M/MPM/30MR) M. H					
SPECIFICATION		DESIGN AND BUILT TO ULC S615						11		SS - 3M/MPM/30MR) MM. H					
								13		SS - 3M/MPM/30MR) M. MRM. H					
								16		SS - 3M/MPM/30MR) M. 30MR) M. H					
QUALITY CONTROL				HEAD TO SHELL JOINT						2M+SS x 6" WIDE INSIDE 5/8" x 10" WIDE OUTSIDE					
BARCOL HARDNESS		MINIMUM		TOP HEAD		SHELL		BOTTOM							
THICKNESS		TOP HEAD		SHELL		BOTTOM		MANWAY BUILT-UP							
INTERIOR CLEAN															
INSPECTED BY:		DATE:		WITNESSED BY:		DATE:									
GENERAL ORIENTATION				DETAILED ORIENTATION											
TANK REQUIREMENTS										NOTE: S = SYNTHETIC VEIL V = C - GLASS VEIL H = 1 1/2 oz CHOPPED STRAND MAT R = WOVEN RYING 24 OZ H = HOT COAT P = PARABEAM PV = PVDF					
LAMINATE LEVEL		STD.		LIFTING LUGS		YES				TANK HANDLING INSTRUCTIONS					
UV PROTECTED		YES		PYRAMID GUSSETS [P]		NO				- TO LIFT TANK, USE CRANE AND SPREADER BAR. - USE ONLY VERTICAL LIFT ON ALL CABLES					
SAVE CUT-OUTS		-													
HEAT TRACING NOZZLES AND RISERS		YES													
DOUBLE WALL FRP TANK										REFERENCE AIR-COR JOB NO. 18-2172-D1					
CERTIFIED FOR PRODUCTION BY:..... DATE:.....				SMR		JUNE 14, 2018		TYPPOS CORRECTED, NOZZLES ADJUSTED							
				SMR		MAY 21, 2018		SUBMITTED FOR REVIEW							
				SMR		MAR 21, 2018		REVISED PER CUSTOMER COMMENTS AS NOTED WITH							
				WVI		FEB 2018		REVISED PER CUSTOMER COMMENTS AS NOTED WITH							
				WVI		MAY 2017		REVISED PER CUSTOMER COMMENTS AS NOTED WITH							
				WVI		OCT 2016		REVISED PER CUSTOMER COMMENTS AS NOTED WITH							
				WVI		-		ISSUED FOR APPROVAL							
BY		DATE						REVISION		NO					
DO NOT SCALE THIS DRAWING USE DIMENSIONS ONLY				PRECISIONEERING   383 HANTRICKY BLVD. SCARBOROUGH ONTARIO CANADA M1P 3P2 Email: sales@PRECISIONEERING.COM											
THE INFORMATION AND KNOW HOW ON THIS DRAWING ARE CONFIDENTIAL AND MAY NOT BE USED OR REPRODUCED WITHOUT WRITTEN PERMISSION OF PRECISIONEERING 				DRAWN		W/ Innumerable		CUSTOMER		Inter Pipeline Propylene Ltd					
				CHECKED		Brock Humphrey		PROJECT		Propane Dehydrogenation Project					
				DATE		MAY 2016		PROJECT LOCATION		Strathcona County, Alberta, Canada					
				SCALE		1" = 1'-0"		EQUIPMENT NAME		Caustic Drain Tank					
				CAD FILE NO.		TH16593-3		EQUIPMENT TAG NO.		10-TA-18005					
CUSTOMER REFERENCE DWG.		PURCHASE ORDER NO.		SHOP ORDER NO.		DRAWING NO.		REV							
A6KV-4-0208-01-00003		A6KV-4-0208-01		TH16593-3		D-16593-3		6							



LEGEND

- 2013 TEST HOLE LOCATION
- SEISMIC CORE PENETRATION TEST (SCPT)
- (SP) STANDPIPE PIEZOMETER (2013)
- 628.1 GROUND WATER ELEVATION (METRES)
- 628.2 m GROUND WATER CONTOURS BASED ON 2013 TEST HOLES

As noted in Thurber's preliminary geotechnical assessment (letter to Williams/Fluor dated July 16, 2013), a preliminary site grade of 631.5 m may be considered based on an average water level of 629.4 m in the main process area as noted in Section 4.3. Water level fluctuations in the order of ± 0.5 m have been recorded at nearby industrial sites, and greater variations may occur during periods of drought or prolonged precipitation events. If a lower risk option is desired that takes this potential variation into account, a site grade elevation of 632.0 m or higher should be selected for the main process areas.



BASE PLAN A6KV-XXX- SK009 REV D PROVIDED BY FLUOR CANADA LTD.



FORT SASKATCHEWAN PROPANE DEHYDRATION
FACILITY GEOTECHNICAL INVESTIGATION

SITE PLAN SHOWING GROUNDWATER
LEVEL ELEVATIONS

DWG No. 19-3500-6-6

DRAWN BY	ML
DESIGNED BY	PSS
APPROVED BY	DJL
SCALE	1:5000
DATE	JULY 2013
FILE No.	19-3500-6



THURBER ENGINEERING LTD.

Coltin Walsh

From: Thomas Aves
Sent: Wednesday, October 6, 2021 9:59 AM
To: Richard Foulkes; Kelty Macleod; Coltin Walsh; Kelsie Sun; Mike Y. Chen
Subject: FW: Tank TA-18005 Caustic drain tank questions
Attachments: A6KV-4-0208-01-00003_GENERAL ARRANGEMENT - 10-TA-18005_R7-AW.pdf

Rick:

Attached is the GA for the Caustic Drain Tank we were just talking about.... for meeting

1. Geotech check on water table - provided information versus actual 2. Check on EHT and whether submergence is an issue 3. Fluor vs Vendor to do the buoyancy level check - Kelty.... this was Mike's question he posed to see what was the best way to deal with this 4. Foundation check on new buoyancy forces 5. Address freezing concerns

-----Original Message-----

From: glenroy.gerald@interpipeline.com <glenroy.gerald@interpipeline.com>
Sent: Tuesday, October 5, 2021 10:57 AM
To: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>; Mike Y. Chen <Mike.Y.Chen@Fluor.com>; Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>
Cc: patrick.wong@interpipeline.com; Ben Santos <Ben.Santos@interpipeline.com>; Shauna Sharp <Shauna.Sharp@Fluor.com>
Subject: [External] RE: Tank TA-18005 Caustic drain tank questions

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Why are we asking the vendor for buoyancy calcs when fluor is responsible for the design and was aware of the Geotechnical data at site? This smell and look like warranty work to me...Pat you need to reach out to Neil and Rick on this.

Thanks,

Glenroy Gerald P.Eng., PMP | InterPipeline Ltd. | Engineering Manager | PDH/CUB – Project Development
Office: 587-475-1154 | Cell: 403.389.4500 | 3200, 215-2nd Street SW, Calgary, AB T2P 1M4

“The person who says it cannot be done should not interrupt the person who is doing it “

-----Original Message-----

From: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>
Sent: Tuesday, October 5, 2021 7:35 AM
To: Mike Y. Chen <Mike.Y.Chen@Fluor.com>; Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>
Cc: Patrick Wong <Patrick.Wong@interpipeline.com>; Ben Santos <Ben.Santos@interpipeline.com>; Glenroy Gerald <Glenroy.Gerald@interpipeline.com>; Shauna Sharp <shauna.sharp@fluor.com>
Subject: RE: Tank TA-18005 Caustic drain tank questions

Good Morning Mike,

Along with the buoyancy calculations, is there any worry about the heat tracing on the inlet elbow, or freezing of the water around the vessel?

Trevor Purcell, P.Eng.

Sr. Site Mechanical Engineer

Inter Pipeline Ltd. | 55463 Range Road 220 | Fort Saskatchewan, AB T8L-4G1

O: 1-587-744-0437 | C: 825-440-8105 | E: Trevor.Purcell@interpipeline.com

<https://nam10.safelinks.protection.outlook.com/?url=http%3A%2F%2Fwww.interpipeline.com%2F&data=04%7C01%7C0ltin.Walsh%40fluor.com%7C522219784d094ff411d908d988e23e35%7C75864cfef26d419cb69dc638695b5533%7C0%7C0%7C637691327569428543%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6Ikl1haWwiLCJXVCi6Mn0%3D%7C1000&data=JJKm%2FH%2F9LiaQ9KD%2BO593fESRjUgWUVvu%2B5F1AVYmhg%3D&reserved=0>

-----Original Message-----

From: Mike Y. Chen <Mike.Y.Chen@Fluor.com>

Sent: Tuesday, September 14, 2021 1:15 PM

To: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>; Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>

Cc: Patrick Wong <Patrick.Wong@interpipeline.com>; Ben Santos <Ben.Santos@interpipeline.com>; Glenroy Gerald <Glenroy.Gerald@interpipeline.com>; Shauna Sharp <shauna.sharp@fluor.com>

Subject: RE: Tank TA-18005 Caustic drain tank questions

Hi Trevor,

Tank vendor is requesting \$6,000 to perform calculations to check if the higher buoyancy forces will have any affect on the tank design as ground level water is high than assumed in original design.

PO for this pkg was closed some times back such that a new PO would be needed in order to proceed, please let us know if this is something you want Fluor to issue or if IPL would like to address directly with vendor.

Thanks,

Mike Chen, P.Eng (AB) | FLUOR CANADA LTD. | Mechanical Engineering | Mike.Y.Chen@fluor.com | +1.403.801.6822

-----Original Message-----

From: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>

Sent: Friday, September 10, 2021 1:30 PM

To: Mike Y. Chen <Mike.Y.Chen@Fluor.com>; Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>

Cc: patrick.wong@interpipeline.com; Ben Santos <Ben.Santos@interpipeline.com>; glenroy.gerald@interpipeline.com

Subject: [External] RE: Tank TA-18005 Caustic drain tank questions

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Good Afternoon Everyone,

The hydrovacing was completed yesterday and it was found that it did no help as the groundwater was coming right back into the inlet nozzle location.

Trevor

Trevor Purcell, P.Eng.

Sr. Site Mechanical Engineer

Inter Pipeline Ltd. | 55463 Range Road 220 | Fort Saskatchewan, AB T8L-4G1

O: 1-587-744-0437 | C: 825-440-8105 | E: Trevor.Purcell@interpipeline.com

<https://nam10.safelinks.protection.outlook.com/?url=http%3A%2F%2Fwww.interpipeline.com%2F&data=04%7C01%7CColtin.Walsh%40fluor.com%7C522219784d094ff411d908d988e23e35%7C75864cfef26d419cb69dc638695b5533%7C0%7C0%7C637691327569438535%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6Ikl1haWwiLCJXVCi6Mn0%3D%7C1000&data=vlLABOsZXSVsQRtfohRIT7oBmWivv89Kk94y6qw9lUs%3D&reserved=0>

-----Original Message-----

From: Mike Y. Chen <Mike.Y.Chen@Fluor.com>

Sent: Wednesday, September 8, 2021 8:30 AM

To: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>; Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>

Cc: Patrick Wong <Patrick.Wong@interpipeline.com>; Ben Santos <Ben.Santos@interpipeline.com>; Glenroy Gerald <Glenroy.Gerald@interpipeline.com>

Subject: RE: Tank TA-18005 Caustic drain tank questions

Hi Trevor,

To give you update, vendor engineering concern is the buoyancy forces and if they might be greater than the ground load assumed in the design. They are checking the original calculations and will reply with comments later this week.

I'm wondering what was result after hydrovac at site? Did the ground water level drop?

Thanks,

Mike Chen, P.Eng (AB) | FLUOR CANADA LTD. | Mechanical Engineering | Mike.Y.Chen@fluor.com | +1.403.801.6822

-----Original Message-----

From: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>

Sent: Tuesday, August 31, 2021 9:41 AM

To: Thomas Aves <Thomas.Aves@Fluor.com>; Kelty Macleod <Kelty.Macleod@Fluor.com>; Mike Y. Chen <Mike.Y.Chen@Fluor.com>

Cc: patrick.wong@interpipeline.com; Ben Santos <Ben.Santos@interpipeline.com>; glenroy.gerald@interpipeline.com

Subject: [External] FW: Tank TA-18005 Caustic drain tank questions

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Good Afternoon,

Inspection of TA-18005 has raised some concerns.

1) Is there an approved drawing from this vendor? Currently in corworx and mycontent we do not have an approved drawing.

2) The ground water design of this vessel is halfway up the drum. As the pictures have shown, the water level is above the inlet piping. The inlet piping is heat traced and insulated.

We are currently hydrovacating out the water to see if the level will still be at the present condition.

My question is, if the ground water level is above design, what are the consequences and how do we rectify this situation? Thanks.

Trevor

Trevor Purcell, P.Eng.

Sr. Site Mechanical Engineer

Inter Pipeline Ltd. | 55463 Range Road 220 | Fort Saskatchewan, AB T8L-4G1

O: 1-587-744-0437 | C: 825-440-8105 | E: Trevor.Purcell@interpipeline.com

<https://nam10.safelinks.protection.outlook.com/?url=http%3A%2F%2Fwww.interpipeline.com%2F&data=04%7C01%7CColtin.Walsh%40fluor.com%7C522219784d094ff411d908d988e23e35%7C75864cfef26d419cb69dc638695b5533%7C0%7C0%7C637691327569438535%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6Ikl1haWwiLCJXVCi6Mn0%3D%7C1000&data=04%7C01%7CColtin.Walsh%40fluor.com%7C522219784d094ff411d908d988e23e35%7C75864cfef26d419cb69dc638695b5533%7C0%7C0%7C637691327569438535%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6Ikl1haWwiLCJXVCi6Mn0%3D%7C1000&sdata=vbLABOsZXSvSQRtfohRIT7oBmWivv89Kk94y6qw9lUs%3D&reserverved=0>

-----Original Message-----

From: Verleen Barry (Consultant) <Verleen.Barry@interpipeline.com>

Sent: Monday, August 30, 2021 1:57 PM

To: Trevor Purcell (Consultant) <Trevor.Purcell@interpipeline.com>

Cc: Barry Woodruff <Barry.Woodruff@interpipeline.com>; Mark Oxenham (Consultant)

<Mark.Oxenham@interpipeline.com>; Justin Cholak (Consultant) <Justin.Cholak@interpipeline.com>

Subject: Tank TA-18005 Caustic drain tank questions

Items identified for review and clarification from TCCC walkdown on 10HP-18A-03 Caustic drain system to TA-18005 Caustic Drain Tank.:

1. Reference red line attachment P&ID PDH-P01-180030A - The inlet pipe at nozzle A02 on drawing and shown in Isometric PDH-P31-WW-43-5702-04. Please update the P&ID to reference the inlet piping going through the corrugated metal pipe and the pipe is U/G. The P&ID shows it is A/G 2. Please reference the first attachment - Precision Engineering Drawing # D-16593-3. It shows the design groundwater table approx. mid section of the tank. See photos 101244 and 101217. This is a view looking into the corrugated metal pipe and showing the inlet pipe with a soft insulation cover. Note the inlet pipe is underwater. Is this acceptable with the upcoming colder temperatures on the way?
3. See photo 102214. This shows nozzles WO1 and A01. The same set up is at the west end of the tank with 3 other nozzles. The protective sleeve around these nozzles will fill with water and freeze. Is this acceptable or should they be covered in and waterproofed.
4. Is there a certified drawing available for attachment 1 from Precision Engineering? The drawing we currently have is from June 14, 2018 - Typos, corrected, nozzles adjusted.

Thank you

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